

摘要：

论文以麒麟2026实测数据回应3D堆叠芯片的散热质疑。数据显示，晶体管密度提升约55%，但相同性能下NPU、GPU功耗分别下降66%和58%。何庭波认为，LogicFolding通过缩短信号传输路径、减少数据搬运能耗，使芯片在“堆得更密”的同时反而降低功耗，也让外界担忧的“ τ 芯片本该过热烧毁”并未成为现实。

堆叠更多晶体管，芯片反而更冷——这一看似反直觉的结论，成为华为半导体负责人何庭波最新论文试图回答的核心问题。

9月4日，何庭波在中科院科技论文预发布平台ChinaXiv发布论文《Huawei's τ Chip Was Supposed to Melt?》（《华为的 τ 芯片本该过热烧毁？》），**通过麒麟2026的实测数据，回应业界对3D堆叠芯片“高密度带来高功耗、高发热”的质疑。**

论文数据显示，麒麟2026晶体管密度较前代提升约55%，但在相同性能下，NPU功耗下降66%，GPU功耗下降58%。**何庭波认为，芯片能耗的关键不只是“计算”，更在于“数据移动”——LogicFolding通过缩短信号传输距离，从源头降低数据搬运能耗。**

据第一财经，今年5月，华为对外发布“ τ 定律”，尝试为后摩尔时代的半导体演进寻找一条新的技术路径。而此次论文更新，也是华为试图向外展示“ τ 芯片”是如何通过重构电路结构、缩短信号传输距离、压缩传输延迟，将“时间维度革新”转化为芯片性能、功耗、温控的突破点，同时回应堆叠芯片“高密度等于高发热”的行业质疑。

对于华为而言， τ 定律能否在功耗、性能、制造和成本之间持续取得平衡，将成为这条技术路线能走多远的关键。

Huawei's τ Chip Was Supposed to Melt?

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Abstract

Heat is the sharpest concern on τ scaling law — the time-scaling principle behind Huawei's folded silicon, named for the delay τ ("Tao") it sets out to shrink. Our Kirin measurements turned that objection upside down. The reason is one every commuter already knows: on a chip, as in a workday, it is the travel that burns the majority of the energy, not the work at the desk.

Index Terms

LogicFolding, τ scaling law, 3D integration, hybrid bonding, low-power design, system-technology co-optimization.

Ask a hundred chip engineers what the most important technology is to the future of the industry and many will answer: 3D integration — stacking dies on dies to boost the amount of silicon you can devote to a system. Ask any experienced chip engineer what will kill 3D integration, and you will almost always hear the same word: heat.

The reasoning feels like a law of nature. Stack active transistors on top of active transistors, cram more into every square millimeter, seal the whole thing inside a compact mobile device, and the outcome seems inevitable. Power turns into heat. Heat trapped in a confined space raises temperature. Temperature past a limit throttles the clock, degrades reliability, and eventually cooks the chip. By this logic, the denser you fold, the hotter you burn — and the harder you fold, the faster you hit the wall.

Huawei cannot wait for that argument to settle itself. Cut off from the tools that let its competitors keep shrinking transistors, the company had to break the cycle — and break it sooner than anyone else. So it stopped scaling space, the linear dimensions of a transistor, and started scaling time: the characteristic delay a signal takes to travel a chip's critical paths. That delay is written as τ — pronounced tau, and rendered in Chinese as Tao (陶), and τ scaling law is a campaign to drive τ down generation after generation [1]. Its flagship technique for compressing that delay is called **LogicFolding**: taking a circuit that used to sprawl across a flat plane and folding it into vertically stacked tiers bridged by tiny vertical links so that signals travel shorter distances.

From the start, τ was conceived as a law of electronic systems rather than a device trick: time can be compressed at every level — device, circuit, chip, and system. LogicFolding is one technique within that framework, working at the circuit and chip levels. The distinction matters, because the industry keeps forgetting it. Soon after the transistor came an invention of equal consequence — the integrated circuit. Yet semiconductor progress is still told as a story about better transistors, as if the circuits integrated from billions of transistors were an afterthought. τ scaling inherits that blind spot: seeing the same transistors inside a τ chip, skeptics conclude that nothing fundamental has changed. The evidence that follows shows why they are wrong.

When LogicFolding was first shown in public — in May 2026, at the IEEE International Symposium on Circuits and Systems — the skepticism in the room was thick. The single sharpest objection pointed toward heat, heat, and more heat.

There is just one problem with that logic. Measurements on real silicon — the Kirin 2026, set to ship this month — showed the opposite. The chip was cooler than its predecessor while comprising 55% more transistors per square millimeter and delivering as much as 66% power reduction on some key tasks. What follows is why that's the case and why LogicFolding and τ scaling law will carry Huawei — and probably others — well into the future.

“芯片最耗电的不是计算，而是数据搬运”

何庭波指出，业界长期更关注晶体管开关产生的计算能耗，却低估了数据在芯片内部传输所消耗的能量。

她以“通勤”作类比：芯片也像上班族，晶体管执行计算需要能量，数据在不同计算模块之间长距离传输同样需要付出能耗。

论文称，在典型智能手机工作负载中，动态功耗约占总功耗的90%。按照动态功耗公式，电容的重要来源并非晶体管本身，而是金属互连。传输距离越长，电容越大，能耗也越高。

何庭波据此认为，在先进工艺节点上，互连电容对能耗的影响已经超过晶体管本身。降低功耗因此不只有缩小晶体管这一条路径，缩短信号传输距离同样重要。

这正是LogicFolding的理论基础。

LogicFolding：把长距离“横向通勤”变成短距离“垂直换乘”

LogicFolding并非简单堆叠芯片，而是重新设计电路布局，将原本横向铺开的平面电路安排至上下两层，并通过混合键合建立高密度垂直互连，把部分长距离水平连接转化为更短的垂直连接。

论文披露，在麒麟2026上，典型核心连线长度缩短约20%，关键路径部分连线最多缩短70%；时钟布线长度缩短28%，时钟缓冲器数量从约4.36万个降至1.9万个。



到麒麟2027，键合间距进一步缩小至1微米，垂直互连点超过1亿个。华为预计未来三年内将键合间距推进至720纳米，垂直互连点超过2亿个。

连线缩短意味着互连电容下降；同时，LogicFolding释放的“时间余量”还可以用于增加并行计算单元，让更多单元以更低频率、更低电压运行，从而进一步降低功耗。

NPU功耗降66%，CPU仍是难点

论文给出的实测数据，成为何庭波回应“ τ 芯片会不会过热”的关键依据。

在相同29 TOPS的AI推理性能下，麒麟2026的NPU时钟频率下降63%，供电电压由0.85V降至0.55V，功耗下降66%，功耗密度下降73%。全速运行时，折叠后的NPU最高可实现70 TOPS，较前代提升141%。

GPU同样表现明显。在相同61帧/秒的性能下，电压降低约200mV，功耗下降58%。

DSP则出现差异。麒麟2026在完成相同工作时功耗下降25%，但由于芯片面积同步缩小40%，功耗密度反而上升24%。何庭波表示，麒麟2027已改善这一问题，DSP功耗下降47%，功耗密度也低于平面设计基准。

CPU性能核心的收益相对有限。在相同性能下，频率仅下降9%，功耗下降41%。原因在于CPU工作负载具有较强的串行特征，难以像NPU、GPU一样通过增加并行计算单元充分利用电压降幅。

何庭波坦言，CPU部分仍需更复杂的设计、EDA工具支持和长期验证，未来三到五年仍需要大量探索。

3D堆叠为何没有更热？关键在“热点”

针对3D堆叠“必然加剧散热”的质疑，何庭波认为，判断散热风险不能只看总发热量，还要看晶体管结温以及热量是否形成局部热点。

LogicFolding主要通过两方面降低热风险：一是在相同性能下减少总功耗，从而降低需要耗散的热量；二是重新安排模块位置，让高功耗模块错位分布，避免热点在垂直方向直接叠加。

麒麟2026也没有将全部电路垂直堆叠，而是重点折叠关键路径，并配合热感知布局。

因此，“ τ 芯片”没有因为3D堆叠而过热，并不意味着堆叠本身解决了散热问题，而是通过缩短信号路径降低能耗，再结合热布局优化，降低整体热负荷和局部热点风险。

τ 定律的边界：时间余量并非“免费性能”

何庭波同时明确， τ 是时间缩放定律，而不是能量缩放定律。

LogicFolding带来的“时间余量”既可以用于降低功耗，也可以用于提升性能。如果将这部分余量全部用于增加计算量，功耗并不会自然下降。因此，麒麟2026的低功耗表现，是设计者主动将部分时间余量用于节能的结果。

这一路线仍面临制造和设计挑战，包括混合键合间距、上下层应力、CMP平坦化与清洗、晶圆翘曲、对准精度，以及更加复杂的EDA设计和验证流程。

何庭波在论文结尾借用西西弗斯的神话称，每一轮芯片迭代都应留下“计算更多、耗能更少”的成果。

对于投资者而言， τ 定律仍有待产业进一步验证。就目前麒麟2026的实测结果看，“ τ 芯片”并未如质疑所言过热：晶体管密度提升55%，NPU功耗反而下降66%。换言之，所谓“本该过热烧毁”，恰恰没有发生——LogicFolding靠的是让数据少走、让能量少浪费。

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